Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice Thematic Assembly Centre
SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010
SEAICE_ANT_PHY_L3_MY_011_018

Issue: 1.1

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	August 2023	All	Creation of the document. The previous document included many Sea Ice TAC products and was split (CMEMS-SI-QUID-011-001to007-009to015-019-V2.17)	Dominique Briand Cédric Prevost Fanny Arduin	C. Wettre
1.1	May 2024	All	Addition of CFOSAT datasets	Cédric Prévost	C. Wettre

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010

Sea ice drift maps are estimated in the Arctic from several scatterometers and radiometers. Single drift fields are combined at the same resolution into a "merged" field, built at two- three- six- and thirty-day lags (depending on the sensor(s) used), during the winter seasons (September until May). ASCAT[1] and merged ASCAT/SSMI[2] ASCAT/QUIKSCAT[3] and also CFOSAT SCAT[4] and merged CFOSAT SCAT/SSMI datasets are ~62.5 km (0.5 degree) spatial resolution while AMSR[5] datasets are ~31.25 km (0.25 degree) spatial resolution. The CFOSAT data are time limited, covering the 2020-2022 period.

SEAICE_ANT_PHY_L3_MY_011_018

Sea ice drift maps are estimated in the Antarctic from AMSR radiometer datasets at ~31.25 km (0.25 degree) spatial resolution since 2003 at two- and three-day lags, during the winter seasons (October until April). CFOSAT SCAT data are also used to build sea ice drift maps at two-, three- and thirty-day lags. CFOSAT SCAT and merged CFOSAT SCAT/SSMI datasets are ~62.5 km (0.5 degree) spatial resolution for the 2020-2022 period.

I.2 Summary of the results

I.2.1 Arctic sea ice drift

SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010

The low resolution Arctic Sea ice drift products have been compared to buoy data (US Interagency Arctic Buoy Program (IABP[6])). The validation shows good correlation between the reference and the product and standard deviation is 7.5 km for 3-day lag drifts and 8.9 km for 6-day lag drifts for both velocity components.

In order to generate the drift maps, the same processing method is applied with the QUIKSCAT, SSMI, ASCAT and CFOSAT data.
Be aware that the original CFOSAT data are still to be assessed by the space Chinese agency.

The medium resolution Arctic Sea ice drift products have also been compared to buoy data (US IABP). The validation (see **Table 1**) shows good correlation between the reference and the product and

1 Advanced Scatterometer sensors developed for the MetOp A, B and C missions.
2 Special Sensor Microwave Imager instrument (passive microwave radiometer) developed for the Defense Meteorological Satellite Program (DMSP).
3 The NASA *QuikSCAT* (Quick Scatterometer) was an Earth observation satellite carrying the SeaWinds scatterometer.
4 *SCAT* is a wind-scatterometer on board the Chinese-French Oceanic SATellite (CFOSAT)
5 Advanced Microwave Scanning Radiometer on board Aqua and GCOM-W satellites.
6 https://www.ncei.noaa.gov/products/international-arctic-buoy-program

standard deviation is 6.2 km for 2-day lag drifts, 6.7 km for 3-day lag drifts, and 8.2 km for 6-day lag drifts for both velocity components.

I.2.2 Antarctic sea ice drift

SEAICE_ANT_PHY_L3_MY_011_018

The Antarctic ice drift time series has been validated with the same method as in the Arctic. It shows good correlation (see **Table 1**) with data from buoys, with a standard deviation of 4.5 km for 2-day lag drift, and 6.0 km for 3-day lag drift with an enhanced standard deviation for the angle, which is 14.6° for 2-day lag drift, and 14.9° for 3-day lag drift. As regard to CFOSAT data, the assessment relies on buoy data to be found for the short 2020-2022 period.

I.3 Estimated Accuracy Numbers

The low resolution Arctic Sea ice drift products (QuikScat, Ascat) have been compared to buoy data (IABP). The validation shows good correlation between the reference and the product and standard deviation is 7.5 km for 3-day lag drifts and 8.9 km for 6-day lag drifts for both velocity components. The medium resolution Arctic Sea ice drift products (AMSR) have also been compared to buoy data (US IABP). The validation (see **Table 1**) shows good correlation between the reference and the product and standard deviation is 6.2 km for 2-day lag drifts, 6.7 km for 3-day lag drifts, and 8.2 km for 6-day lag drifts for both velocity components.

Table 1: *Arctic and Antarctic Sea Ice drift products standard deviations compared against buoys*

Product Name	Type	Variable	Unit	Description
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010	Sea Ice drift	northward_sea_ice_velocity eastward_sea_ice_velocity	$m.s^{-1}$	Compared against buoys: Low resolution (SSMI / QuickSCAT / ASCAT) STD: 7.5 km for 3 day lags STD: 8.9 km for 6 day lags Medium resolution (AMSR) STD: 6.2 km/39.9° for 2 day lags STD: 6.7 km/35.2° for 3 day lags STD: 8.2 km/26.8° for 6 day lags
SEAICE_ANT_PHY_L3_MY_011_018	Sea Ice drift	northward_sea_ice_velocity eastward_sea_ice_velocity	$m.s^{-1}$	Compared against buoys: Medium resolution (AMSR) STD: 4.5 km/14.6° for 2 day lags STD: 6.0 km/14.9° for 3 day lags

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: CERSAT IFREMER, France

Production system name: SIW-IFREMER-BREST-FR

This section is presenting the used satellite data and the applied algorithms to infer the products.

The products SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010 and SEAICE_ANT_PHY_L3_MY_011_018 contain datasets calculated from different Brightness Temperature (BT) acquired from different sensors.

II.1 Data collection

The first step is to collect the original satellite data from different data providers listed in Table 2 below:

***Table 2**. Input data provider per Copernicus Marine Service products and sensor used.*

Copernicus Marine Service products	Sensor	Provider
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010	SSM/I	National Snow and Ice Data Center (NSIDC) https://n5eil01u.ecs.nsidc.org/PM/NSIDC-0080.002/
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010 SEAICE_ANT_PHY_L3_MY_011_018	AMSR2	Bremen University Arctic : https://data.seaice.uni-bremen.de/amsr2/tb_daygrid_swath/n6250/ Antarctic : https://data.seaice.uni-bremen.de/amsr2/tb_daygrid_swath/s6250/
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010	ASCAT	Eumetsat ASCSZR1B
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010	QUICKSCAT	(No longer operational)
SEAICE_ARC_SEAICE_REP_OBSERVATIONS_011_010 SEAICE_ANT_PHY_L3_MY_011_018	CFOSAT SCAT	CNES

II.2 Per sensor drift processing

II.2.1 Sea ice concentration

In order to calculate the ice drift, a first step is necessary to compute the sea ice concentration. This is done using SSMI data and specifically the ARTIST Sea Ice (ASI) algorithm developed at the University of Bremen and adapted at IFREMER (see [3] for more details).

II.3 Sea ice drifts

II.3.1 Preprocessing

A preprocessing of Brightness temperature is applied in order to deal with the “blind-area” around the North Pole which is assimilated to a land area, adjusted daily to each map, and expanded by a three pixel-wide band to account for the median filter applied (see [3] for more details).

II.3.2 Drift processing

The drift algorithm and data flagging are identical for all datasets. The first stage is the Laplacian computation followed by the maximum correlation search. A preliminary coarse drift validation is performed using an interpolated ECMWF[7] wind field averaged over the different time resolution of the datasets. Then local consistency controls are performed using neighbouring drift vectors. Depending on the sensor used, the vertically and horizontally polarised brightness temperature maps are processed independently in order to perform a cross-validation check at the merging stage.

A more detailed methodology can be found in [4] whereas AMSR datasets specificity is detailed in [5] and SSMI in [3].

7 European Centre for Medium-Range Weather Forecasts.

III VALIDATION FRAMEWORK

III.1 Arctic ice drift time series

The assessed parameter is sea Ice drift velocity.

For the Arctic, the sea ice drift has been compared with buoys from IABP (US). U and V components have been analysed through five winters from 1999 to 2004 (magnitude and direction). The same assessment method was applied for each dataset, however the winter period used for comparison was not always the same and varied depending on the dataset (see table 3 and see [1] for more details. For AMSR – one winter 2002-2003).

The validation results (see Table 3) show good correlation between the reference and the product and standard deviation, for example for AMSR, is 6.2 km for 2-day lag drifts, 6.7 km for 3-day lag drifts, and 8.2 km for 6-day lag drifts for both velocity components.

Table 3: *Arctic Sea Ice drift products standard deviations compared against buoys (Girard-Ardhuin and Ezraty, 2012).*

rms drift magnitude difference (km)	rms drift difference (cm s^{-1})	rms direction difference (°)	day lag (nb of days)	product	period
7.5 **7.5[e]** **8.9** **9.1[e]**	**2.91** **2.90[e]** **1.72** **1.76[e]**	**39.2** **29.6[e]** **29.6** **24.3[e]**	**3** **3** **6** **6**	**Merged SSM/I & QuikSCAT**	**five winters**
8.1 **10.8**	**3.14** **2.08**	**35.1** **27.0**	**3** **6**	**Merged SSM/I & QuikSCAT**	**one winter**
6.2 **6.7** **8.2**	**3.61** **2.60** **1.59**	**39.9** **35.2** **26.8**	**2** **3** **6**	**Merged AMSR-E H & AMSR-E V**	

III.2 Antarctic ice drift time series

Following the same assessment method, the Antarctic product has been compared with several buoys during the 2004-2021 period (April to October) and the results analysed in terms of magnitude and direction. For this assessment, we referred to 3 major programs: PANGAEA[8], IPAB[9] and IGC[10] for a total of 77 buoys (see **Figure 1**).

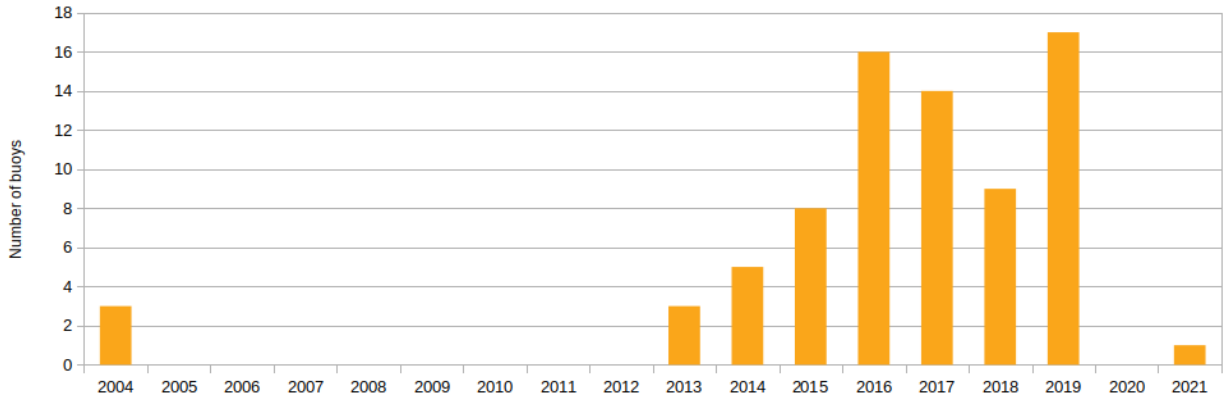

***Figure 1**: Time series of the number of buoys available (Y) in the Antarctic that was used for the Antarctic product validation (Pangaea, IPAB and IGC buoys) over the years (X).*

8 Data Publisher for Earth & Environmental Science (https://www.pangaea.de/about/).

9 International Program for Antarctic Buoys (https://cmr.earthdata.nasa.gov/search/concepts/C1214313568-AU_AADC)

10 Ice Guiding Central buoys, part of broader programs (IABP, IOEB, ...)

IV VALIDATION RESULTS

IV.1 Arctic ice drift time series

Radiometers and scatterometers have been extensively used for Arctic Sea ice drift estimation, the validation of single sensor drift products with buoys shows good accuracy. The merged drifts are compared with IABP buoy drifts at three- and six-day lags. Results are presented in Table 3 through drift magnitude and direction parameters (also done with east/north components). The velocity parameter has been added in Table 3 for information, due to our method, the constant half-pixel minimum displacement embedded in the original measurement is then erroneously accounted for.

***Table 3**. Results of the Comparisons (Root Mean Square – RMS-) Between Satellite Drift Vectors (Merged Drifts at Low and Medium Resolutions) and Buoy Drifts: Standard Deviation for Sea Ice Drift Difference (Magnitude and Velocity) and for Direction Difference, Number of Day Lag, Product, and Period. Excluding Small Drifts (Less Than One Pixel).*

rms drift magnitude difference (km)	rms drift difference (cm s^{-1})	rms direction difference (°)	day lag (nb of days)	product	period
7.5 7.5[e] 8.9 9.1[e]	2.91 2.90[e] 1.72 1.76[e]	39.2 29.6[e] 29.6 24.3[e]	3 3 6 6	Merged SSM/I & QuikSCAT	five winters
8.1 10.8	3.14 2.08	35.1 27.0	3 6	Merged SSM/I & QuikSCAT	one winter
6.2 6.7 8.2	3.61 2.60 1.59	39.9 35.2 26.8	2 3 6	Merged AMSR-E H & AMSR-E V	

As explained and described in [1], the results are good as compared with other product validations (see also [6]).

Note that it is very difficult to compare and to conclude from validation results, mainly because of the different buoy data sets, methodologies, areas, time periods, etc. Moreover, the results have been reported with several scales (drift magnitude and direction, east and north components, X- and Y-axis components, and ice velocity). The drift estimation is a mean displacement and not a velocity, and usually, the estimated displacements are converted into ice speed in order to compare the quality of the drift magnitude estimated at different day lags or at various pixel sizes with other works, even though this transformation is nonlinear. This problem of the comparison of results has been discussed and highlighted with more details in [2] and in [6].

IV.1.1 Low resolution

The standard deviation of the difference to buoys is 7.5 km/39.2° for 3 days lag drift and 8.9 km/29.6° for 6 days lag drifts. Figure 2 shows the angle difference between the product and buoys as a function of merged drift magnitudes: the angle difference sharply decreases (smaller than 45°) for drift magnitudes higher than 40 km (about three pixels). This result in drift direction is comparable to previous validation results. The six-day lag is better for small drift magnitude estimation, and consequently, the angle data have a better resolution with a standard deviation of about 30° whereas for the 6 day lag product, it is about 40°. This is a very good point highlighted in [1].

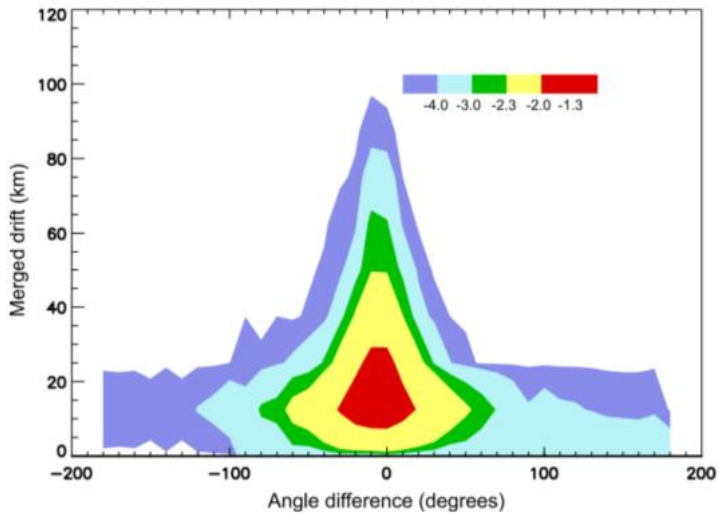

***Figure 2**: Angle difference between satellite merged drift vectors and buoy drift vectors as a function of the merged drift magnitude. The time lag is three days. The comparison spans over five winters. Colors represent the percentage of occurrence expressed in logarithmic scale.*

IV.1.2 Medium resolution

For the medium resolution product, the standard deviation of the difference to buoys is 6.2 km/39.9° for 2days lags drift, 6.7 km/35.2° for 3 days lags drift and 8.2 km/26.8° for 6 days lag drifts.

A comprehensive presentation of the Arctic product validation can be found in [1].

IV.2 Antarctic ice drift time series

IV.2.1 Medium resolution

As presented in Figure 3, in the Antarctic region, the AMSR drifts are very close to the buoy drifts as more than 95% of AMSR-buoy distance measurements are less than 10 km. The standard deviation of the difference to buoys is 4.5 km/14.6° for two-day lag drifts and 6.0 km/14.9° for three-day lag drifts.

As presented in Figure 3, in the Antarctic region, the AMSR drifts are very close to the buoy drifts as to buoy measurements because more than 95% of data are gathered between 0 and 10 km. The standard deviation of the difference to buoys is 4.5 km/14.6° for two-day lag drifts and 6.0 km/14.9° for three-day lag drifts.

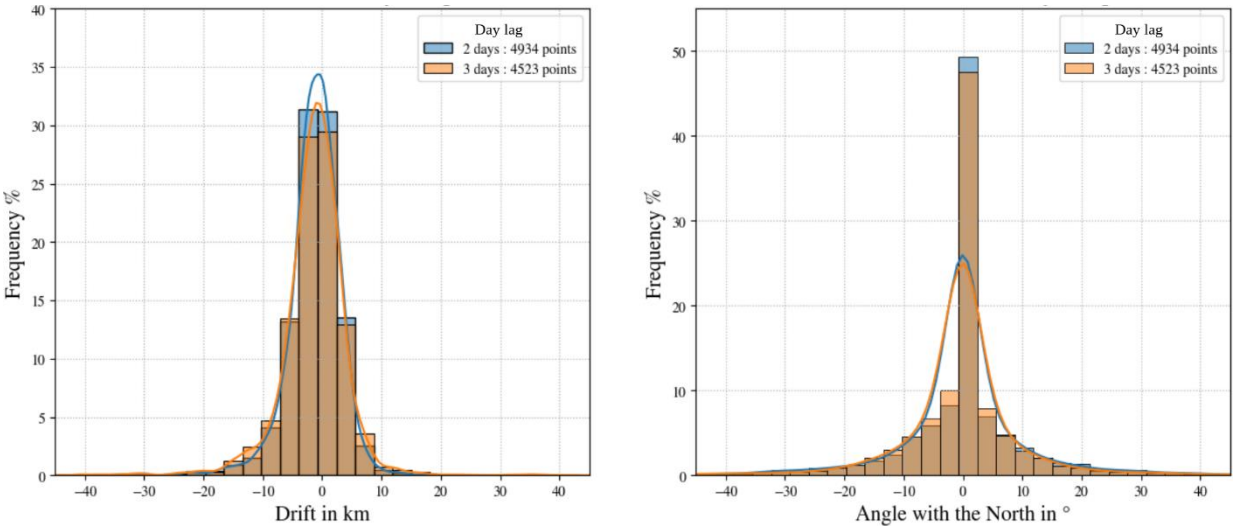

***Figure 3**: Residual data (AMSR minus buoy) with 2 and 3 day lags during the 2004-2021 winter periods. AMSR-buoy displacement differences (left), AMSR-buoy direction differences (right).*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version	Other
01/07/2015	Entry into operation of Sea Ice Drift processing chain for CMEMS. (Arctic and Antarctic products)	1.0.0	
31/05/2024	Update of the document with new datasets	1.1.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

The previous QUID version was aggregating information from several CMEMS ice products. The document has been split to get product dedicated document. New datasets have been added issued from CFOSAT scatterometer sensor. These datasets are time limited, covering the 2020-2022 period. .

VII REFERENCES

[1] Girard-Ardhuin, F., R. Ezraty, 2012: Enhanced Arctic sea ice drift estimation merging radiometer and scatterometer data. IEEE Trans. Geosci. Remote Sensing, vol. 50 (7), July 2012, pp 2639-2648. Doi : 10.1109/TGRS.2012.2184124. http://archimer.ifremer.fr/doc/00088/19895/17729.pdf

[2] Lavergne, T., S. Eastwood, Z. Teffah, H. Schyberg and L. A. Breivik, 2010. Sea ice motion from low-resolution satellite sensors: An alternative method and its validation in the Arctic, *J. Geophys. Res.*, vol. 115, pp. C10032, 2010.

[3] Ezraty, R., F. Girard-Ardhuin, J.F. Piollé, 2007. Arctic & Antarctic sea ice concentration and arctic sea ice drift estimated from special sensor microwave data. User's manual V2.1. February 2007. ftp://ftp.ifremer.fr/ifremer/cersat/products/gridded/psi-drift/documentation/ssmi.pdf

[4] Ezraty, R., F. Girard-Ardhuin and J.F. Piollé, 2007. Sea ice drift in the central arctic estimated from seawinds/quikscat backscatter maps. User's manual V2.2. February 2007.

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[6] Sumata, H., T. Lavergne, F. Girard-Ardhuin, N. Kimura, M.A. Tschudi, F. Kauker, M. Karcher, R. Gerdes, 2014: An intercomparison of Arctic ice drift products to deduce uncertainties estimates. *J. Geophys. Res. Ocean,* vol. 119, July 2014. DOI: 10.1002/2013JC009724.